1) $y = \dfrac{x^3}{2(x+1)^2}$

2) $y = \dfrac{\sqrt{|x-1|}}{x^2+1}$ (no y'')

3) $y = x - 2\log(2x+1) + 1$

4) $y = (x^2+2x-1)\,e^{-2x}$

5) $y = \cos x - \cos^2 x$ (limitatamente a $[0, 2\pi]$)

6) $y = \operatorname{arctg} \dfrac{1}{x^2}$

7) $y = \operatorname{Sh} x - \operatorname{Ch} x$

8) $y = \log(x^2 - x^3)$

9) $y = \log \dfrac{x^2}{x-1}$

10) $y = \dfrac{x^2-1}{e^x}$

11) $y = \cos x \cdot e^{\frac{1}{2}\operatorname{sen} x}$ (no y'')

12) $y = \sqrt{x^2-3x+2} - x$ (no y'')

①

: $y = \dfrac{x^3}{2(x+1)^2}$ CE: $x+1 \neq 0$

$x \neq -1$

$(-\infty, -1)(-1, +\infty)$

Intersezioni con gli assi

$y(0)=0$ e $y=0$ solo per $x=0$

Segno

$y \geq 0$ per $x \geq 0$ (il denominatore sempre ≥ 0)

limiti

$\lim_{x \to \pm\infty} y = \pm\infty$ è un infinito di primo ordine

$\dfrac{\infty_3}{\infty_2}$ ∞_1

poiché ∞ 1° ordine mi chiedo se c'è As. Obl

$\lim_{x \to \pm\infty} \dfrac{y}{x} = \dfrac{1}{2}$ (m) infatti $\dfrac{y}{x} = \dfrac{x^2}{2x^2+4x+2}$

$\lim_{x \to \pm\infty} \left(y - \frac{1}{2}x\right) = -1$ (q)

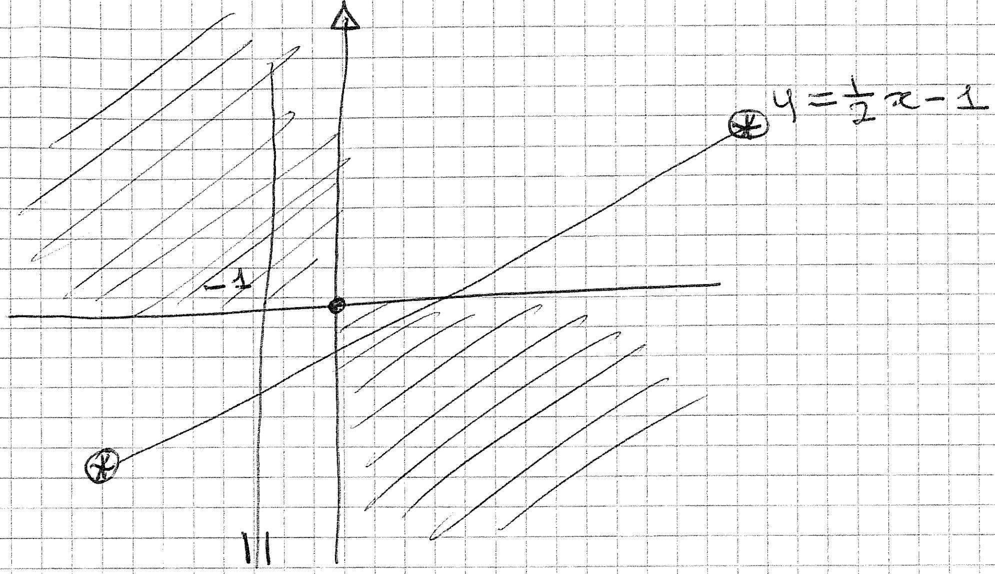
infatti

$y - \frac{1}{2}x = \dots = \dfrac{-2x^2-x}{2x^2+4x+2} \rightarrow -1$

$\Rightarrow$ $y = \frac{1}{2}x - 1$ As. Obl per $x \to \pm\infty$

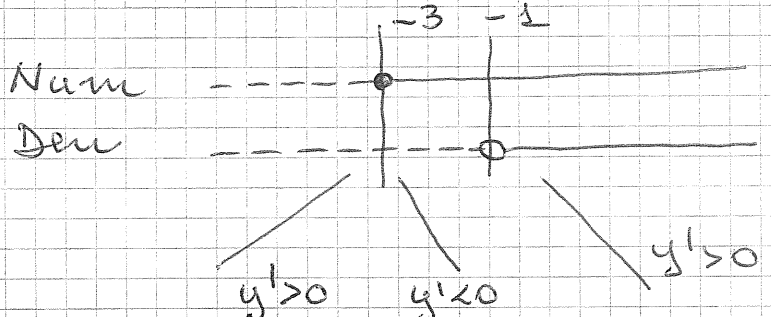
$\lim_{x \to -1} y = -\infty$ infatti $\dfrac{-1}{2 \cdot 0^+} \Rightarrow -\infty$

$x = -1$ As. Verticale



Crescere e decrescere

$$y' = \frac{3x^2(x+1)^{\cancel{2}} - 2x^3\cancel{(x+1)}}{2(x+1)^{\cancel{4}\,3}} = \frac{3x^3+3x^2-2x^3}{2(x+1)^3} = \frac{x^2(x+3)}{2(x+1)^3}$$



$x=-3$, $y(-3) = -\frac{27}{8}$ c'è un massimo

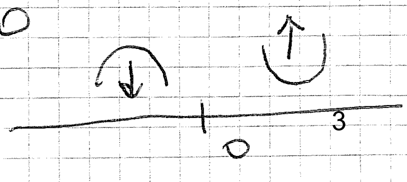
$y'=0$ per $x=0$ ma in $U(0)$ si mantiene positiva $\Rightarrow$ funz ha flesso a tangente orizzontale

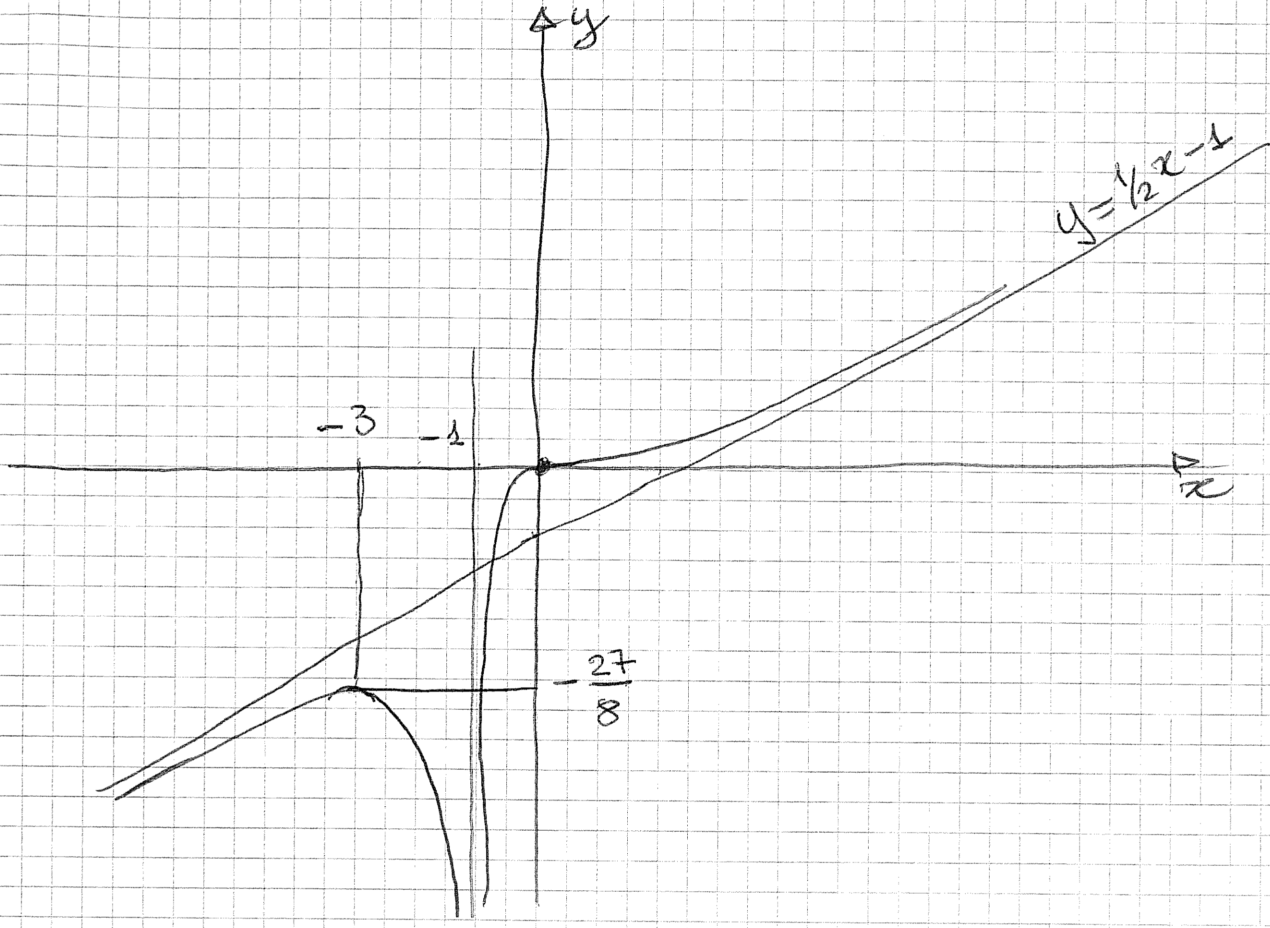
Concavità

$$y'' = \frac{(3x^2+6x)(x+1)^{\cancel{3}} - (x^3+3x^2)\,3\,\cancel{(x+1)}^2}{2(x+1)^{\cancel{6}\,4}} = \dots$$

$$= \frac{3x}{(x+1)^4} \geq 0 \quad \text{per } x \geq 0$$

concavità:





y
$y = \frac{1}{2}x - 1$
x
-3
-1
$-\frac{27}{8}$

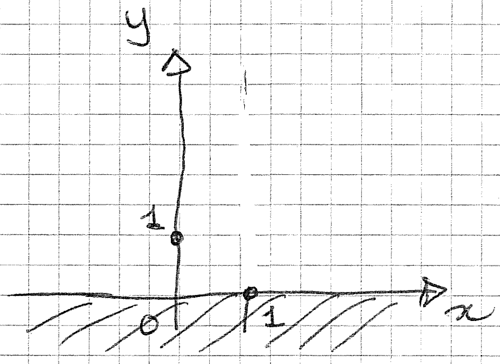
2

$$y = \frac{\sqrt{|x-1|}}{x^2+1} \qquad CE: (-\infty, +\infty)$$

Intersezione assi

$x=0 \rightarrow y(0)=1$

$y=0 \rightarrow x=1$



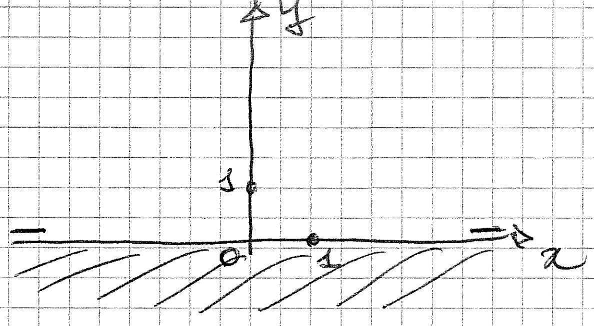
Segno

y sempre ≥ 0, $=0$ per $x=1$

limiti

$$\lim_{x \to \pm\infty} y = 0^+ \qquad \frac{\infty_{1/2}}{\infty_2}$$

$\Rightarrow$ l'asse x è asintoto orizz. a $\pm\infty$



Crescere e decrescere e concavità

Conviene ora "liberarci" del modulo

(1) se $x \geq 1$

$$y_1 = \frac{\sqrt{x-1}}{x^2+1}$$

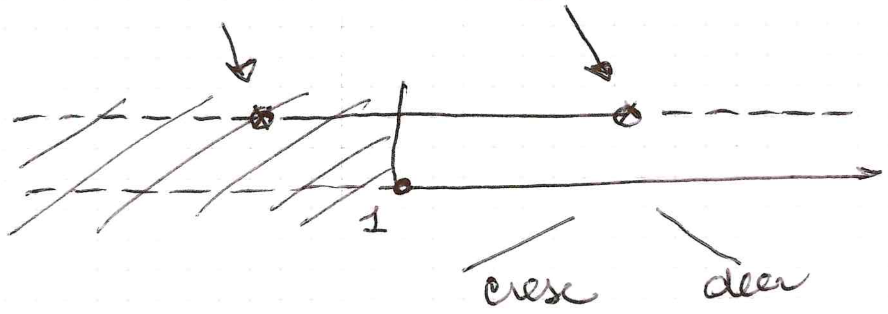
(2) se $x < 1$

$$y_2 = \frac{\sqrt{1-x}}{x^2+1}$$

① $$y_1' = \frac{\frac{1}{2\sqrt{x-1}}(x^2+1) - 2x\sqrt{x-1}}{(x^2+1)^2}$$

$$= -\frac{3x^2-4x-1}{2\sqrt{x-1}\,(x^2+1)^2} \geq 0 \qquad x = \frac{2\pm\sqrt{4+3}}{3}$$

$$y_1' \geq 0 \quad \left(\frac{2-\sqrt{7}}{3} \leq x \leq \frac{2+\sqrt{7}}{3}\right) \cap x \geq 1$$



$\Rightarrow$ in $x = \frac{2+\sqrt{7}}{3}$ c'è un massimo

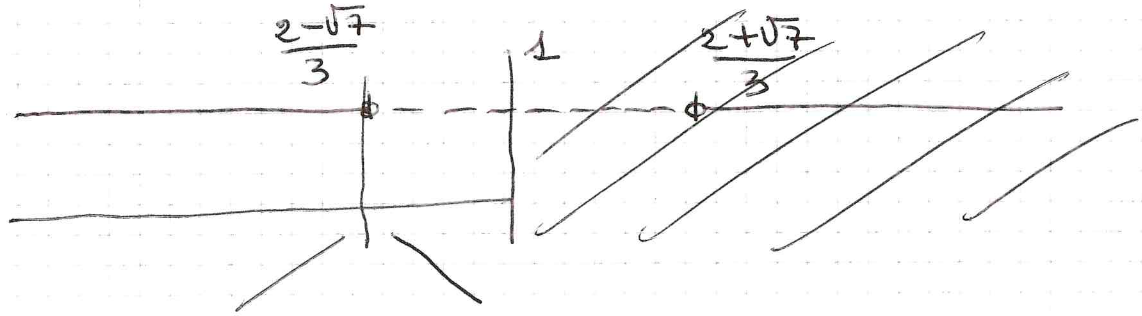
osserviamo che y_1' non è definita in $x=1$

analizziamo il comportamento della sua pendenza

per $x \to 1^+$: $\lim_{x\to 1^+} y_1' = +\infty$ $\Rightarrow$ tangente verticale

② $$y_2' = \frac{3x^2-4x-1}{2\sqrt{1-x}\,(x^2+1)^2}$$

$y_2' \geq 0$ per $3x^2-4x-1 \geq 0$



in $x = \frac{2-\sqrt{7}}{3}$ c'è un massimo

$$\lim_{x \to 1^-} y_2' = -\infty$$ (studio della pendenza in $x=1$ da sinistra)

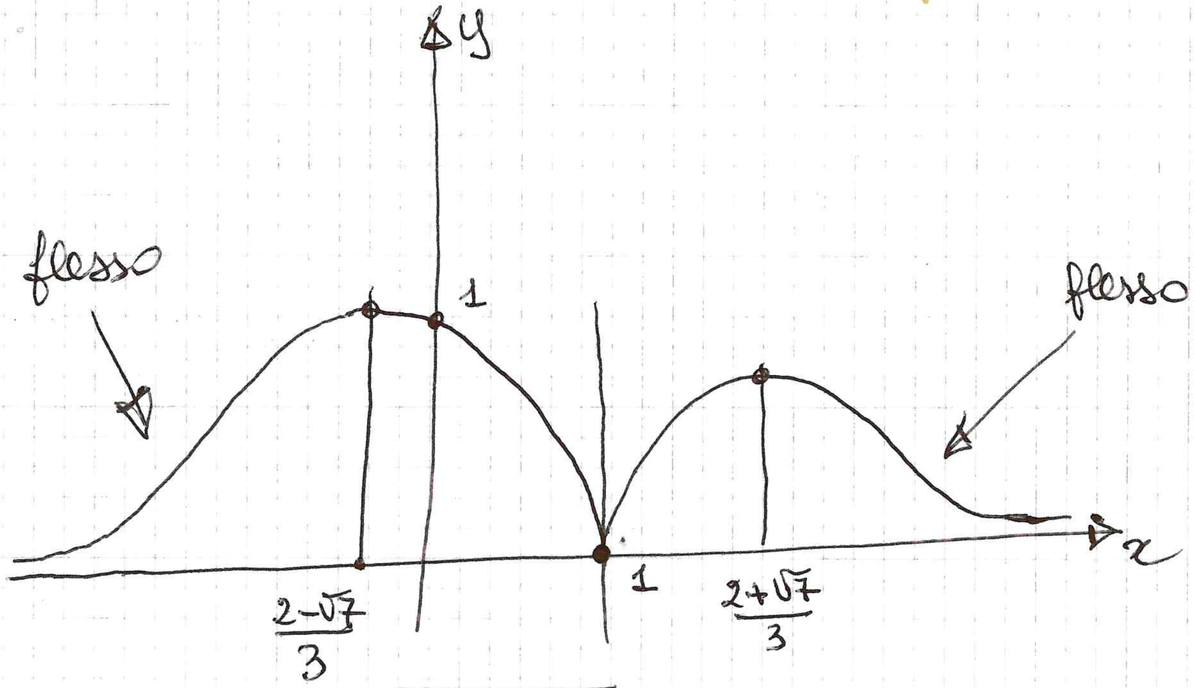
$\Rightarrow$ in $x=1$ c'è una cuspide

Per quanto riguarda la concavità osserviamo che sarà verso l'alto in $U(-\infty)$ e $U(+\infty)$ perché c'è un asintoto orizzontale ($y=0$) e la funzione è positiva.

Inoltre localmente, ove ci sono i massimi sarà verso il basso

Dall'analisi del crescere/decrescere concludiamo che la cuspide è verso il basso.

Senza lo studio della derivata seconda deduciamo che devono essere presenti almeno due flessi.



$$y\left(\frac{2-\sqrt{7}}{3}\right) = \frac{\sqrt{1-\frac{2-\sqrt{7}}{3}}}{\left(\frac{2-\sqrt{7}}{3}+1\right)^2} \quad ; \quad y\left(\frac{2+\sqrt{7}}{3}\right) = \frac{\sqrt{\frac{2+\sqrt{7}}{3}-1}}{\left(\frac{2+\sqrt{7}}{3}+1\right)^2}$$

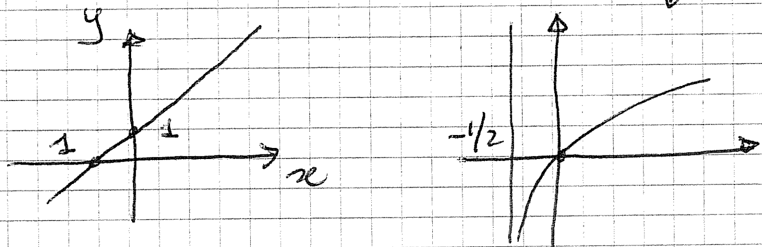
$y = x - 2\log(2x+1) + 1$ CE: $x > -1/2$

$(-1/2, +\infty)$

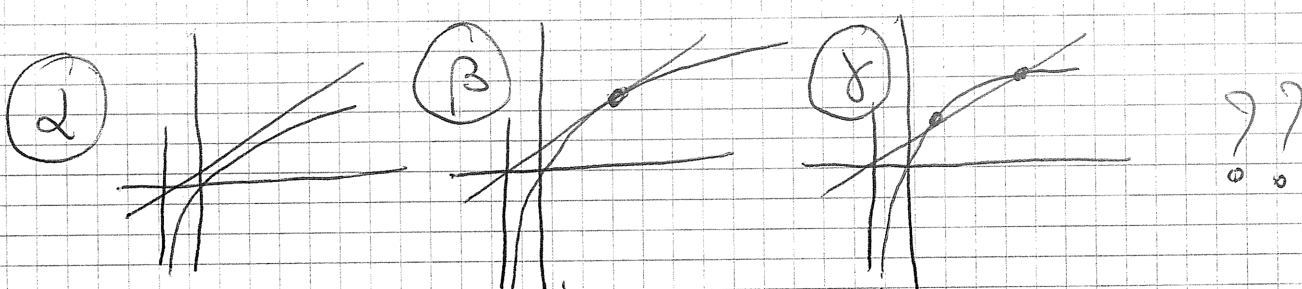
<u>Intersezione con gli assi e segno</u>

$y(0) = 1$

$y = 0$ per $x + 1 = 2\log(2x+1)$



e se le sovrapponiamo?



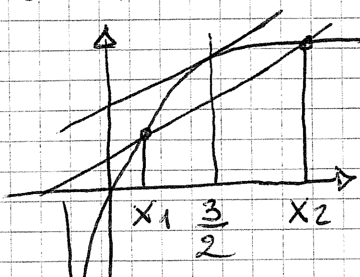
$y = 2\log(2x+1)$ $y' = 2\,\frac{2}{2x+1}$

$y' = 1$ (come la pendenza della retta) se $\frac{4}{2x+1} = 1$

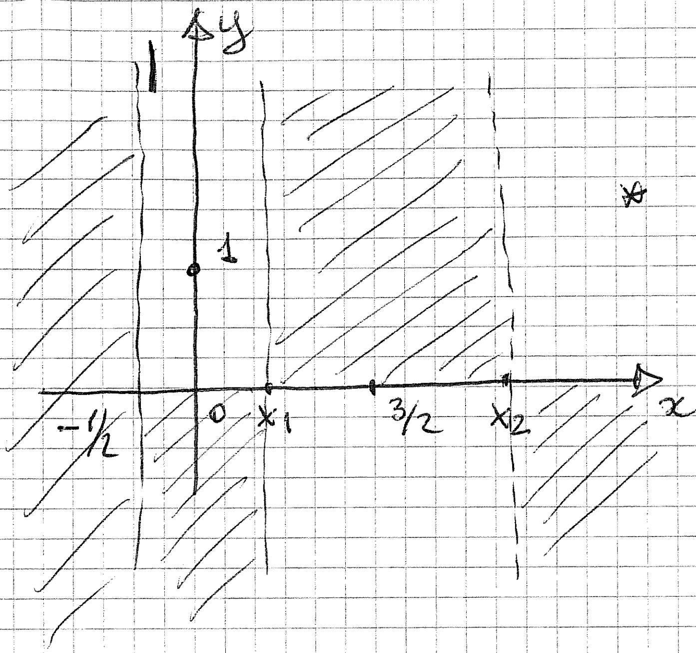
$4 = 2x + 1$ $x = \frac{3}{2}$

$\text{retta}\left(\frac{3}{2}\right) = \frac{5}{2}$ $\left(2\log(2x+1)\right)\left(3/2\right) = \overbrace{2\log 4}^{\approx 2.77} < 5/2$

$\Rightarrow$



caso γ tang alla log // alla retta al di sopra della retta



la funzione data è positiva su $(-1/2, x_1)$ e su $(x_2, +\infty)$

x_1 ed x_2 sono valori approx

$$0 < x_1 < 3/2$$

$$x_2 > 3/2$$

limiti

$$\lim_{x \to -\frac{1}{2}} y = +\infty \qquad -1/2 \underbrace{-\log 0}_{-(-\infty)} + 1$$

$$\lim_{x \to +\infty} y = +\infty \qquad +\infty_{pot} \cancel{- \infty_{log}}$$

è un infinito di I° ordine indaghiamo per ∃ asintoto obliquo

$$\lim_{x \to +\infty} \frac{y}{x} = 1 \quad (m) \qquad \text{infatti} \quad \frac{x - 2\log(2x+1) + 1}{x} =$$

$$1 - 2\underbrace{\frac{\log(2x+1)}{x}}_{\frac{\infty_{log}}{\infty_{pot}} \to 0} + \underbrace{\frac{1}{x}}_{0}$$

$$\lim_{x \to +\infty} (y - x) = -\infty \qquad (\cancel{x} - 2\log(2x+1) + 1) \cancel{- x}$$

$$\nexists\, q \qquad -2\log(+\infty)$$

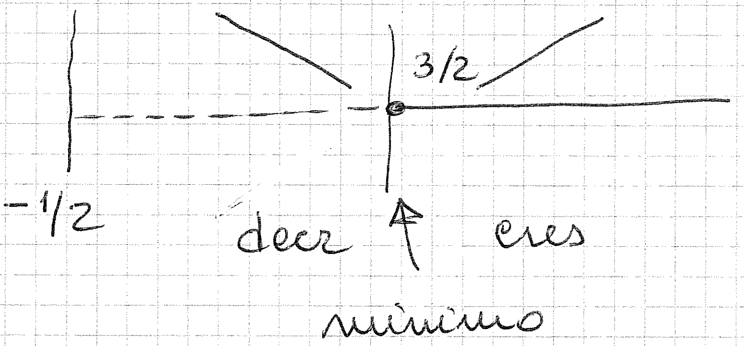
$$-\infty$$

non c'è asintoto obliquo

crescere e decrescere

$$y' = 1 - 2\,\frac{2}{2x+1} = \frac{2x+1-4}{2x+1} = \frac{2x-3}{2x+1}$$

$y' \geq 0 \quad 2x-3 \geq 0 \quad$ (il den > 0 in CE)

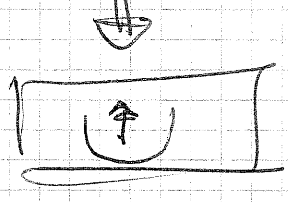


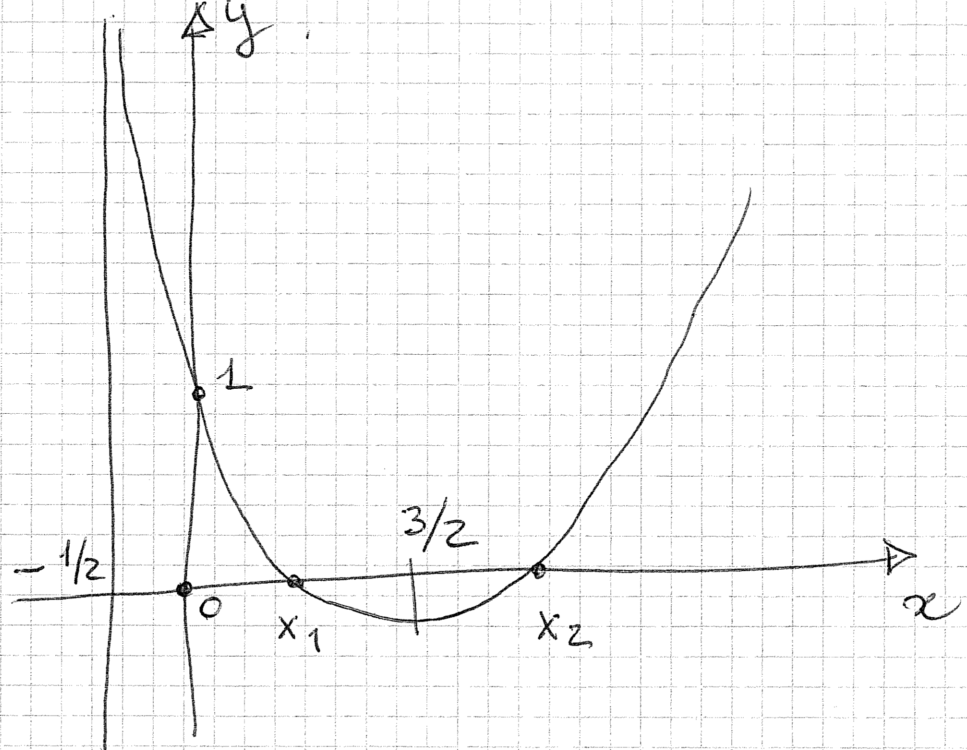
$x = 3/2 \qquad y(3/2) = \frac{3}{2} - 2\log 4 + 1 = \underbrace{\frac{5}{2} - 2\log 4}_{\text{quota del minimo}} < 0$

concavità

$$y'' = \frac{2(2x+1) - 2(2x-3)}{(2x+1)^2} = \frac{8}{(2x+1)^2} > 0 \text{ sempre}$$

$\Downarrow$





(4)

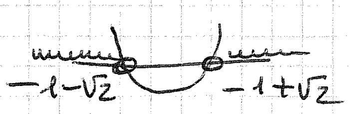
$y = (x^2+2x-1)\, e^{-2x}$ CE $(-\infty, +\infty)$

Segno ed intersezione assi

$y(0) = -1$

$y \geq 0$ se $x^2+2x-1 \geq 0$ $x = -1 \pm \sqrt{2}$

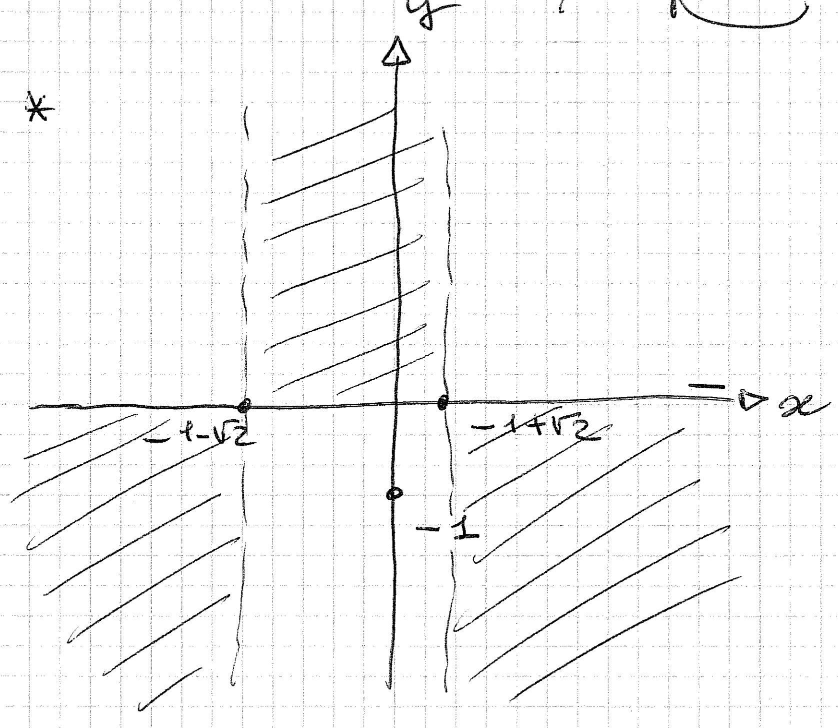
$x \leq -1-\sqrt{2}$, $x \geq -1+\sqrt{2}$



limiti

$\lim_{x \to -\infty} y = +\infty$ $\quad \underset{\substack{\uparrow \\ \text{tipo } x^2}}{\infty} \cdot \underset{\substack{\uparrow \\ +\infty \text{ esp}}}{e^{+\infty}} \Rightarrow +\infty$

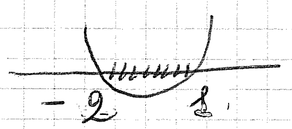
$\lim_{x \to +\infty} y = 0^+$ $\quad \cancel{\underset{\substack{\uparrow \\ \text{tipo } x^2}}{\infty}} \cdot \underbrace{e^{-\infty}}_{0^+ \text{ esp}} \Rightarrow 0^+$

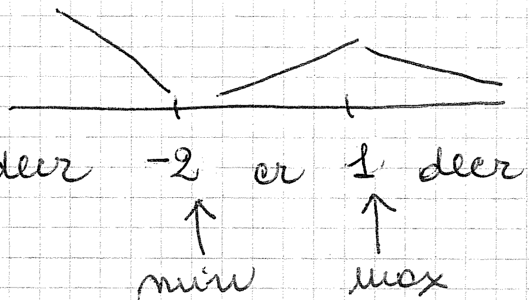


Crescere e decrescere

$$y' = (2x+2)e^{-2x} - 2e^{-2x}(x^2+2x-1)$$
$$= -2e^{-2x}(x^2+x-2)$$

$y' \geq 0 \qquad x^2+x-2 \leq 0 \qquad x = \dfrac{-1 \pm \sqrt{1+8}}{2} = \left\langle \begin{matrix} 1 \\ -2 \end{matrix} \right.$



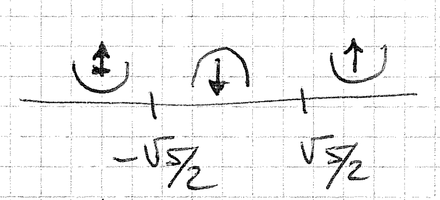


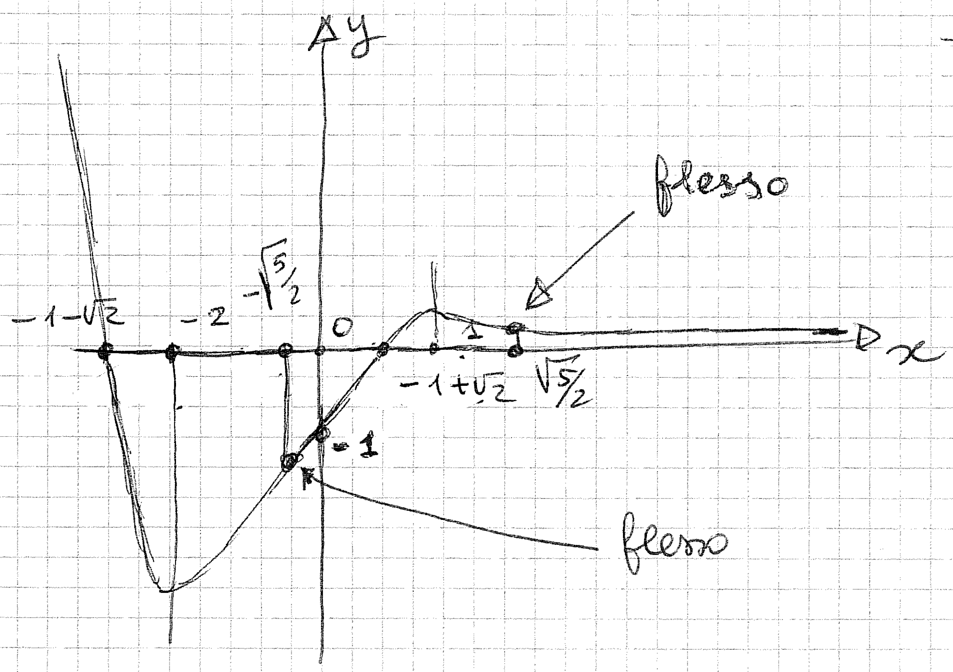
$y(-2) = -e^4$ quota del minimo

$y(1) = 2e^{-2}$ quota del massimo

Concavità

$y'' = 2e^{-2x}(2x^2-5) \geq 0 \qquad x \leq -\sqrt{5/2} \quad x \geq \sqrt{\frac{5}{2}}$





$y = \cos x - \cos^2 x$

non ci sono limitazioni CE $(-\infty, +\infty)$ ma le studiamo su $[0, 2\pi]$ è periodica di periodo 2π

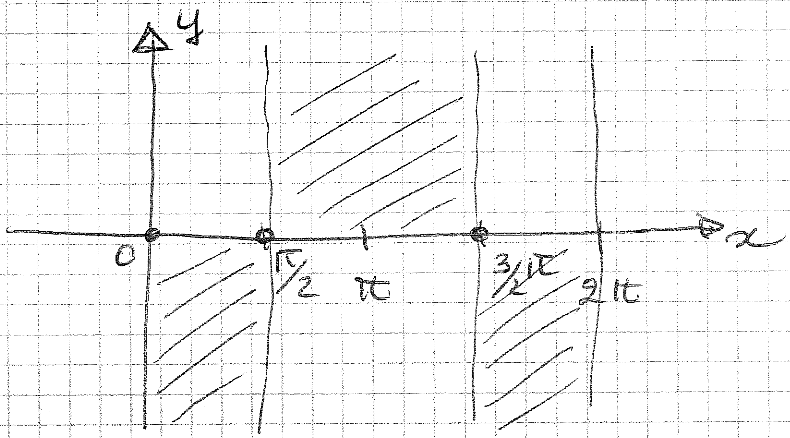
$y(0) = y(2\pi) = 0$

$y \geq 0$ se

$\cos x \, (1 - \cos x) \geq 0$

$\uparrow$ positivo nel 1° e 4° quadrante, 0 per $x = \pi/2$ e $3/2\,\pi$

sempre >0, ad eccezione di $x=0$, $x = 2\pi$ dove $\cos x = 1$



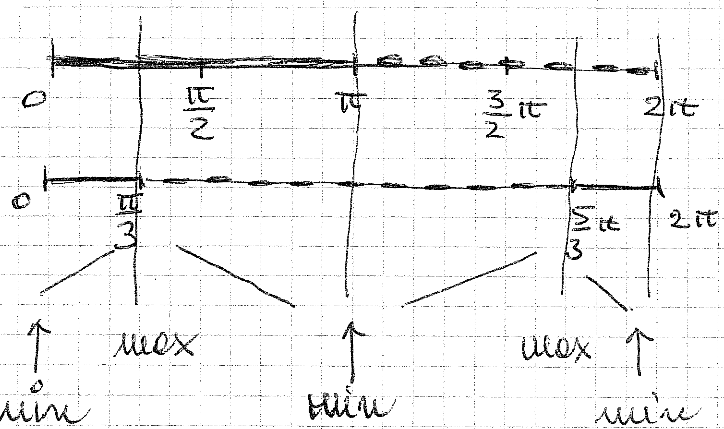
ovviamente non si calcolano limiti perché i punti di bordo (o frontiera) del CE appartengono al CE e quindi in quei punti abbiamo calcolato il valore della funzione

Crescenza e decrescenza

$$y' = -\operatorname{sen} x - 2\cos x(-\operatorname{sen} x)$$
$$= -\operatorname{sen} x \,(1 - 2\cos x)$$
$$= \operatorname{sen} x \,(2\cos x - 1)$$

$y' \geq 0$ senx

$2\cos x - 1$



quote dei minimi

$$y(0) = y(2\pi) = 0 \qquad y(\pi) = -2$$

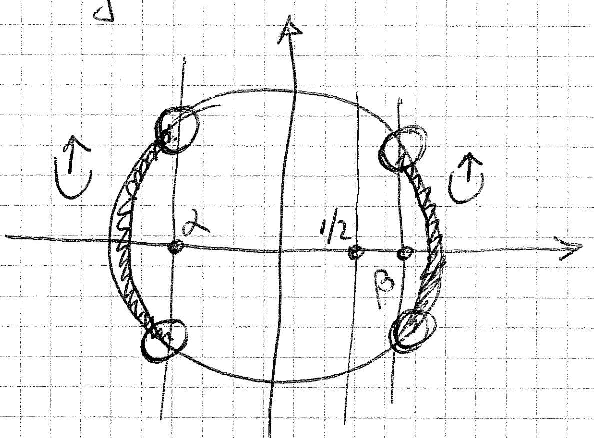
quote dei massimi

$$y(\pi/3) = \frac{1}{4} \qquad y\left(\frac{5}{3}\pi\right) = 1/4$$

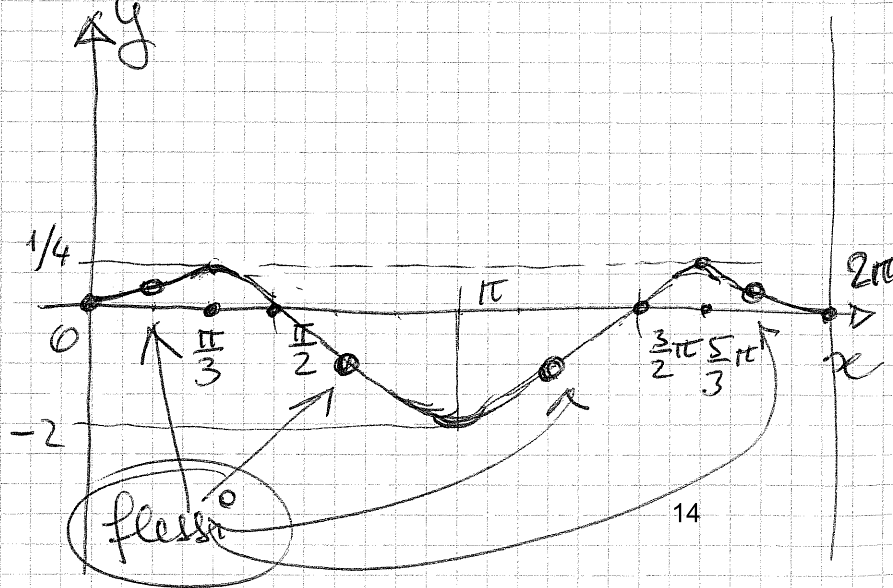
<u>concavità</u>

$$\begin{aligned} y'' &= \cos x(2\cos x - 1) + \operatorname{sen} x(-2\operatorname{sen} x) \\ &= 2\cos^2 x - \cos x - 2\operatorname{sen}^2 x \\ &= 4\cos^2 x - \cos x - 2 \end{aligned}$$

$$y'' \geq 0 \qquad \cos x \leq \underbrace{\frac{1-\sqrt{33}}{8}}_{\alpha} \qquad \cos x \geq \underbrace{\frac{1+\sqrt{33}}{8}}_{\beta}$$



i punti indicati sono corrispondenti a flessi



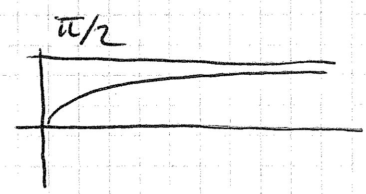
$y = \operatorname{arctg} \frac{1}{x^2}$ CE: $x \neq 0$ (6)

$y(-x) = y(x)$ è pari

la studiamo su $(0, +\infty)$

$y(0)$ no! perché $x = 0 \notin$ CE

$y = 0$ se $\operatorname{arctg} \frac{1}{x^2} = 0$ ma

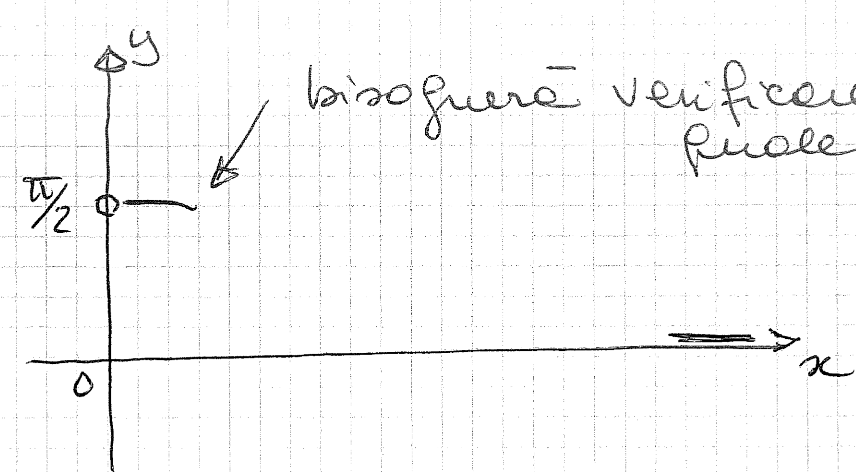


$\Rightarrow$ $y = 0$ mai

in $(0, +\infty)$ y è sempre positiva

$\lim_{x \to 0^+} y = \frac{\pi}{2}$ infatti $\operatorname{arctg}(+\infty)$

$\lim_{x \to +\infty} y = 0^+$ infatti $\operatorname{arctg}(0^+)$



bisognerà verificare con quale pendenza

Crescere e decrescere

$$y' = \frac{1}{1 + \left(\frac{1}{x^2}\right)^2} \cdot \left(-2 \frac{1}{x^3}\right) = \frac{-2x}{1 + x^4}$$

y' è sempre negativa su $(0, +\infty)$

$\Rightarrow$ y decresce

$$\lim_{x\to 0^+} y' = 0^-$$

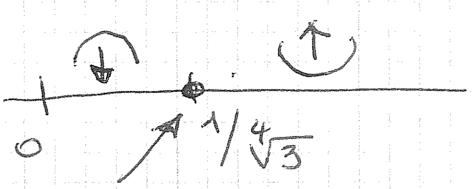
la funzione si attesta verso il punto $(0, \pi/2)$ a tangente orizzontale

concavità

$$y'' = -2\,\frac{1+x^4 - x\cdot 4x^3}{(1+x^4)^2} = \frac{3x^4-1}{(1+x^4)^2}$$

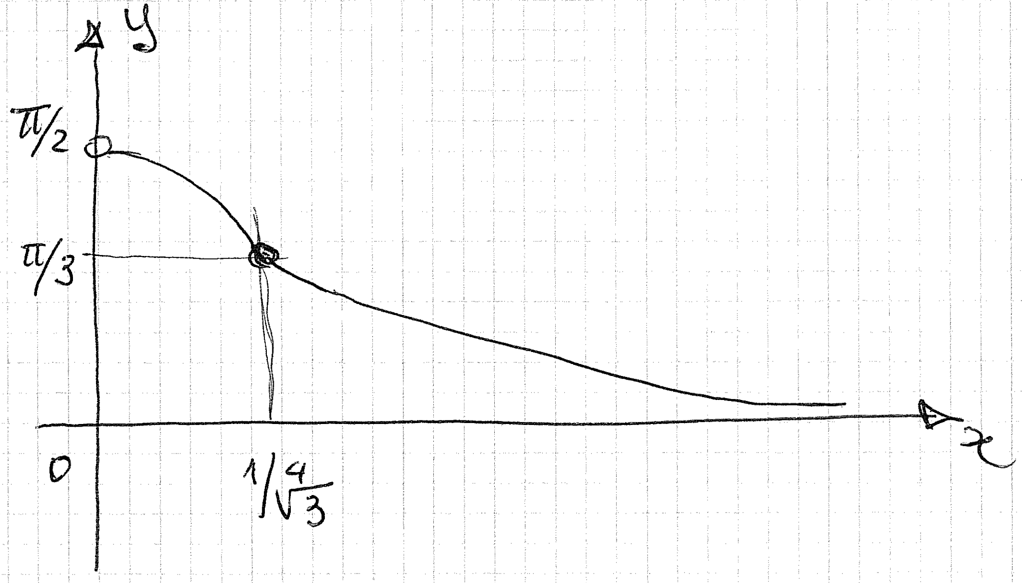
$$y'' \geq 0 \qquad \underbrace{\cancel{(\sqrt{3}x^2+1)}}_{\substack{>0\\ \text{sempre}}}(\sqrt{3}x^2-1)\geq 0$$

$\cancel{x \leq -\sqrt{\frac{1}{\sqrt{3}}}}$ non ci interessa $\qquad x \geq \sqrt{\frac{1}{\sqrt{3}}} = \frac{1}{\sqrt[4]{3}}$



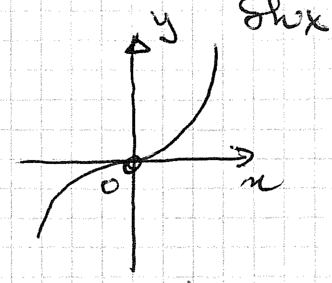
flesso

$$y\left(\frac{1}{\sqrt[4]{3}}\right) = \text{artg}\sqrt{3} = \frac{\pi}{3}$$

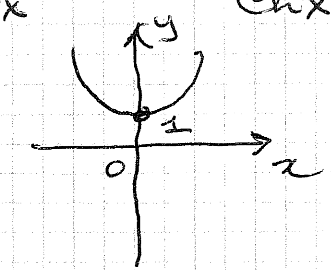


$y = \mathrm{Sh}\,x - \mathrm{Ch}\,x$

Shx



Chx

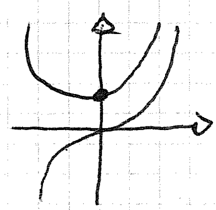


CE: $(-\infty, +\infty)$

$(\mathrm{Sh}\,x)' = \mathrm{Ch}\,x$ $\quad$ $(\mathrm{Ch}\,x)' = \mathrm{Sh}\,x$

$y(0) =$

$y = 0\,?$ poiché



y è sempre negativa

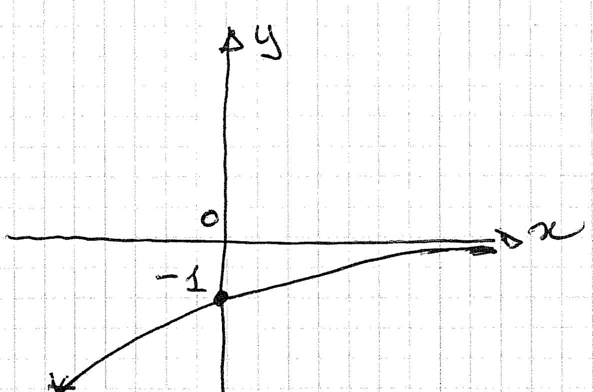
comunque facendo i conti:

$$\frac{\cancel{e^{x}} - e^{-x}}{2} - \frac{\cancel{e^{x}} + e^{-x}}{2} = -e^{-x} < 0$$

$\lim\limits_{x \to -\infty} y = -\infty$ $\qquad$ $-\infty - (+\infty)$

$\lim\limits_{x \to +\infty} y = 0^-$ $\qquad$ $\dfrac{\cancel{e^{x}} - e^{-x}}{2} - \dfrac{\cancel{e^{x}} + e^{-x}}{2} = -e^{-x}$

$-e^{-\infty}$ $\qquad$ -0^+



questo è un $-\infty$ di tipo esponenziale

$y' = \mathrm{Ch}\,x - \mathrm{Sh}\,x$

$y' = -y$ quindi > 0 sempre

y è sempre crescente

$y'' = \mathrm{Sh}\,x - \mathrm{Ch}\,x$

$y'' = y$ quindi < 0 sempre

y ha $\cap$ sempre

$y = \log(x^2 - x^3)$ CE: $x^2 - x^3 > 0$ (8)

$x^2(1-x) > 0$

$x < 1$ e $x \neq 0$

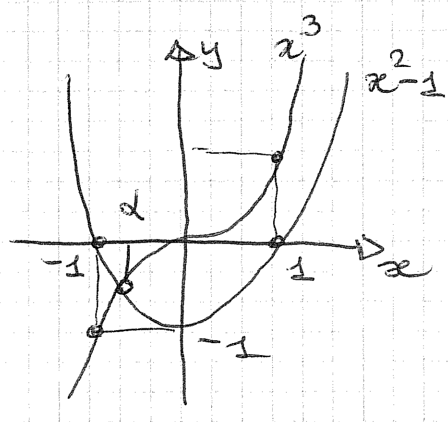
$(-\infty, 0)(0, 1)$

$y \geq 0$ se $x^2 - x^3 \geq 1$

$x^2 - 1 \geq x^3$

$x \leq \alpha$

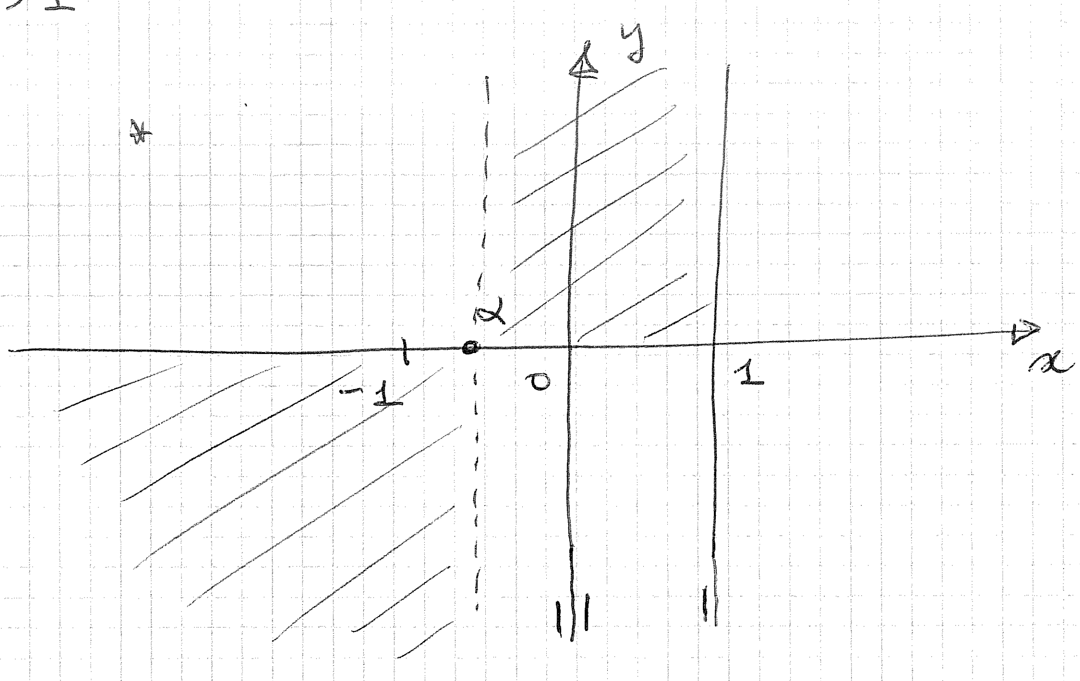
ove $-1 < \alpha < 0$



$\lim_{x \to -\infty} y = +\infty$ di tipo log infatti $\underline{\log}((-\infty)^2 - (-\infty)^3)$

$\lim_{x \to 0} y = -\infty$ è $\log 0^+$

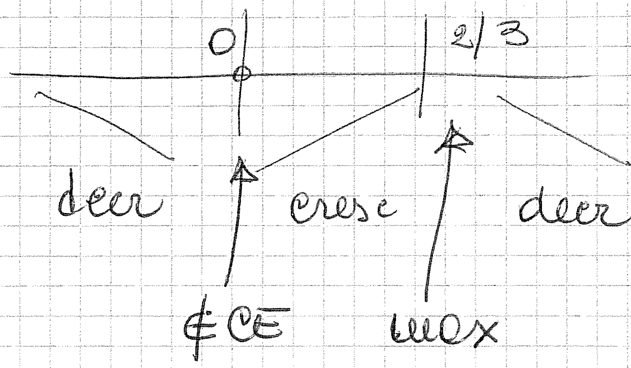
$\lim_{x \to 1^-} y = -\infty$ è $\log 0^+$



$$y' = \frac{1}{x^2 - x^3} \cdot (2x - 3x^2) = \frac{x(2-3x)}{x^2 - x^3}$$

$y' \geq 0$ $x(2-3x) \geq 0$ trascuro il den perché è > 0 in CE

$y' \geq 0 \qquad 0 < x \leq 2/3$

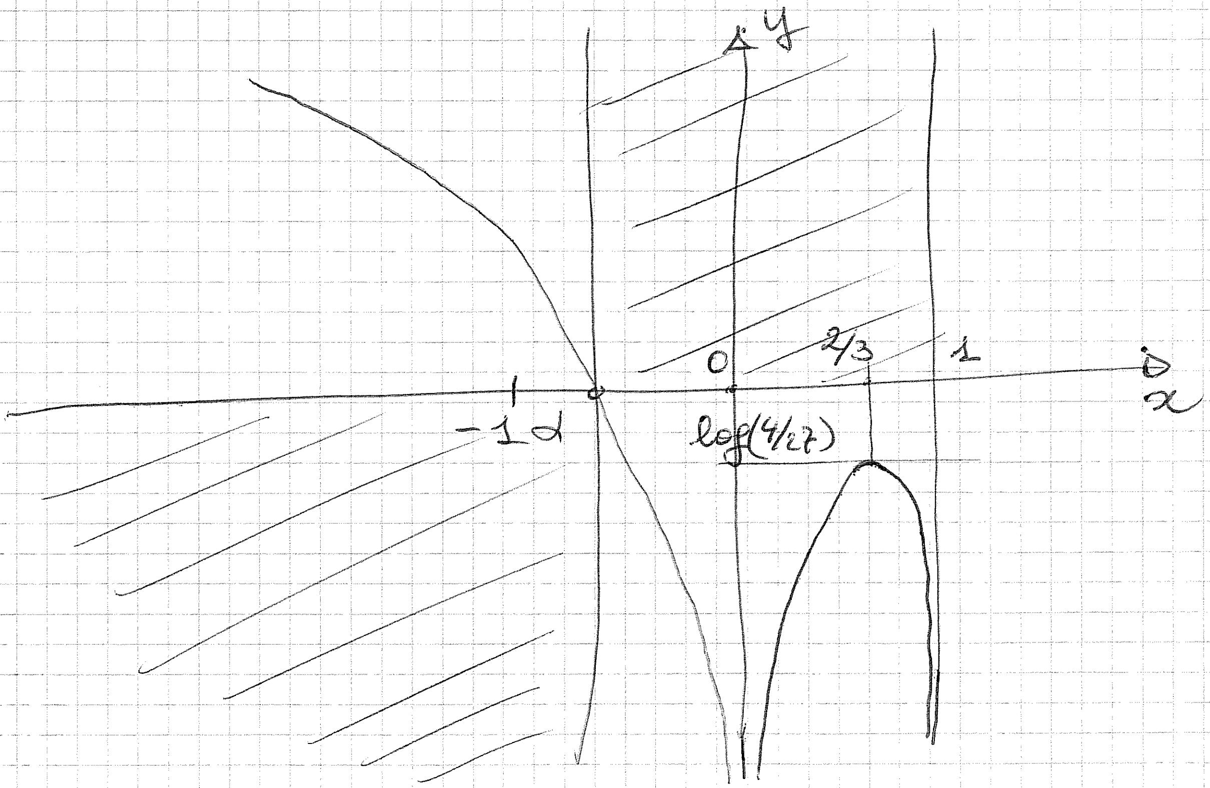


$y'(2/3) = \log\left(\frac{4}{9} - \frac{8}{27}\right) = \log \frac{4}{27} \quad (<0)$ punto del massimo

$$y'' = \left(\frac{2-3x}{x-x^2}\right)' = \frac{-(3x^2-4x+2)}{(x-x^2)^2}$$

$y'' \geq 0$ se $3x^2-4x+2 \leq 0 \qquad \frac{\Delta}{4} = 4-6 < 0$

y'' è sempre negativa



$y = \log \dfrac{x^2}{x-1}$ CE: $\dfrac{x^2}{x-1} > 0$ $x > 1$ $x \neq 0$ non serve

$(1, +\infty)$

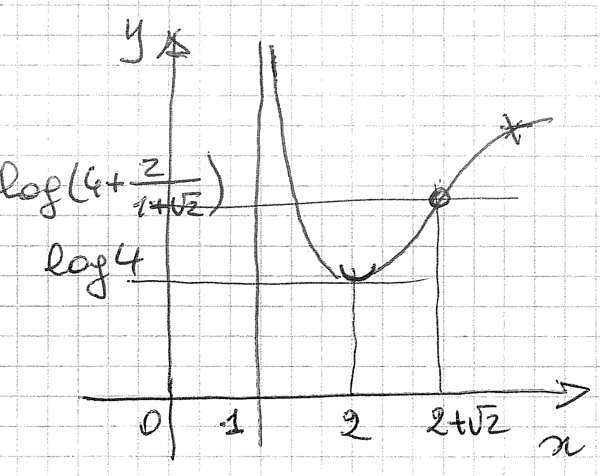
$y \geq 0$ $\dfrac{x^2}{x-1} \geq 1$ $x^2 \geq x - 1$ (il den è >0 nel CE)

$x^2 - x + 1 \geq 0$ $\Delta = -3$

quindi y è sempre positiva

$\lim_{x \to 1} y = +\infty$ infatti $\log \frac{1}{0}$ $\log(+\infty)$

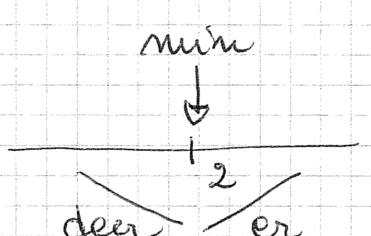
$\lim_{x \to +\infty} y = +\infty$ infatti $\log \dfrac{\infty_2}{\cancel{\infty_1}}$ $\log(+\infty)$



$y' = \dfrac{1}{\frac{x^2}{x-1}} \cdot \dfrac{2x(x-1) - x^2}{(x-1)^2}$

$= \dfrac{x-2}{x(x-1)}$

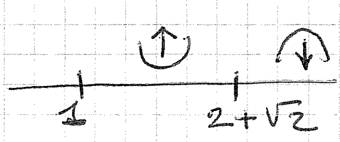
$y' \geq 0$ $x \geq 2$ se den >0 in CE



in $x = 2$ c'è un minimo la cui quota è $\log 4$

$y'' = \dfrac{x^2 - x - (x-2)(2x-1)}{x^2(x-1)^2} = -\dfrac{x^2 - 4x + 2}{x^2(x-1)^2}$ flesso

$y'' \geq 0$ $x^2 - 4x + 2 \leq 0$ $2 - \sqrt{2} \leq x \leq 2 + \sqrt{2}$



$\uparrow < 1$ $y(2+\sqrt{2}) = \log\left(4 + \frac{2}{1+\sqrt{2}}\right)$

$$y = \frac{x^2-1}{e^x} \qquad CE \quad (-\infty, +\infty)$$

10

$$y(0) = -1$$

$y \geq 0$ per $x^2 - 1 \geq 0 \qquad x \leq -1, \; x \geq 1$

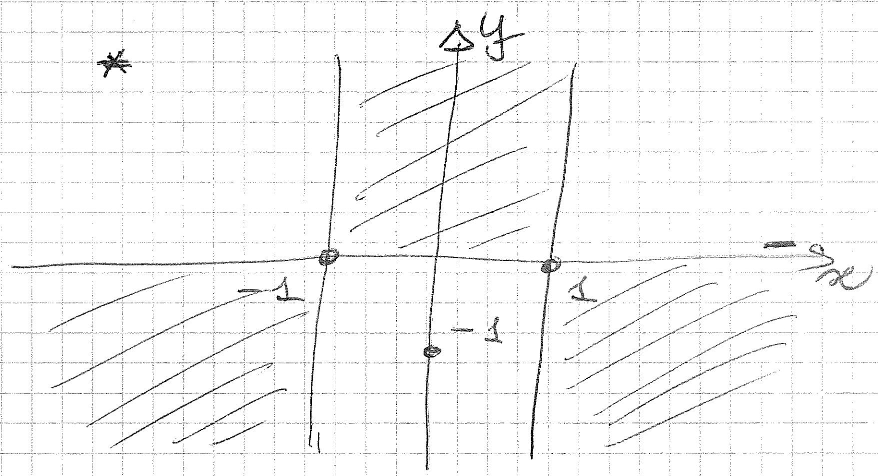
$$\lim_{x \to -\infty} y = +\infty \qquad \frac{(-\infty)^2-1}{e^{-\infty}} \qquad \frac{+\infty}{0^+} \qquad +\infty$$

di ordine $> 2 \Rightarrow$ NO AS. OBL.

$$\lim_{x \to +\infty} y = 0^+$$

$\frac{\infty_2}{e^{+\infty}}$ prevale l'∞ dell'esponenziale

*



$$y' = \frac{2x e^x - (x^2-1)e^x}{e^{2x}} = -\frac{x^2-2x-1}{e^x}$$

$y' \geq 0 \qquad x^2 - 2x - 1 \leq 0 \qquad x = 1 \pm \sqrt{2}$

min Max

$1-\sqrt{2}$ $\quad$ $1+\sqrt{2}$

decr cresce decr

$$y(1-\sqrt{2}) = \frac{(1-\sqrt{2})^2-1}{e^x} = \frac{2(1-\sqrt{2})}{e^{1-\sqrt{2}}} < 0$$

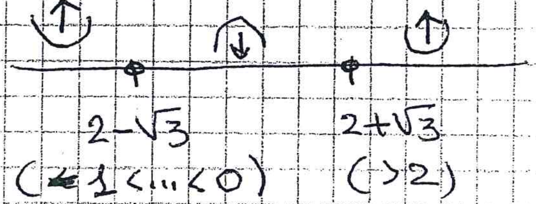
anche senza calcolatrice, con un po' di calcoli e approssimazioni, si riesce a vedere che la quota del minimo è <-1

$$y'' = -\frac{(2x-2)e^x - e^x(x^2-2x-1)}{e^{2x}} =$$

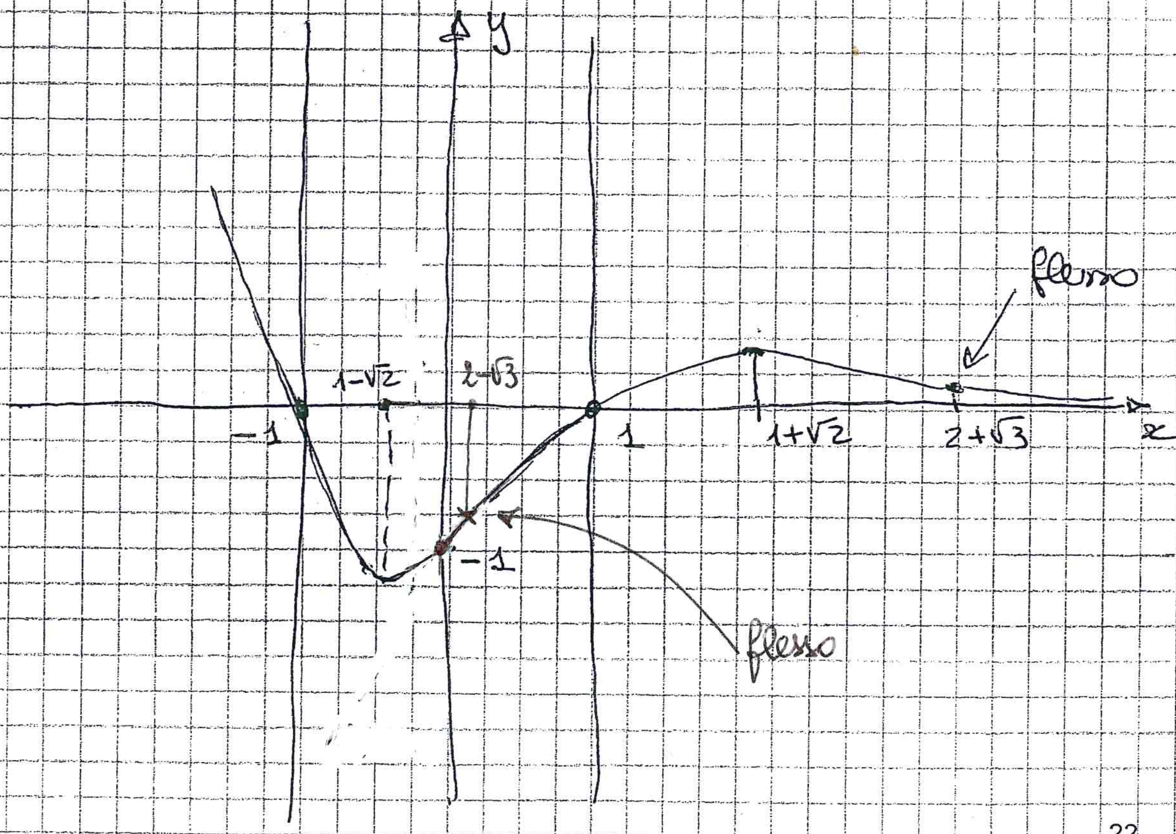
$$= \frac{x^2-4x+1}{e^x}$$

$y'' \geq 0 \quad x^2-4x+1 \geq 0$

$x \leq 2-\sqrt{3} \qquad x \geq 2+\sqrt{3}$



$2-\sqrt{3} > 1-\sqrt{2}$



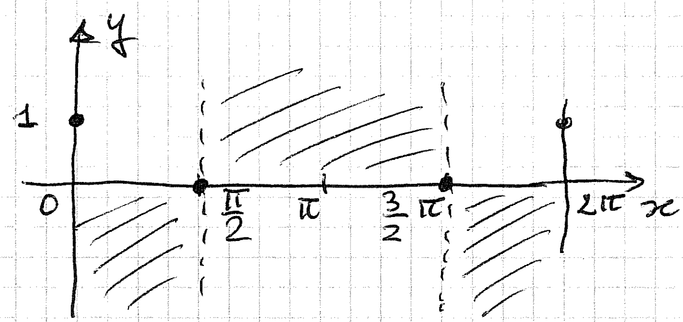
$y = \cos x \, e^{\frac{1}{2} \operatorname{sen} x}$ (11) $(-\infty, +\infty)$ $\boxed{\text{no } y''}$

periodica di periodo 2π $\Rightarrow$ $[0, 2\pi]$

$y(0) = y(2\pi) = \cos 0 \cdot e^{\frac{1}{2}\operatorname{sen} 0} = 1$

$y = 0$ se $\cos x = 0 \Rightarrow x = \frac{\pi}{2}$ e $x = \frac{3}{2}\pi$

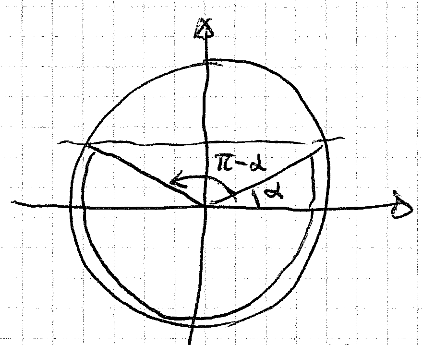
$y > 0$ $\cos x > 0$ nel 1° e 4° quadrante

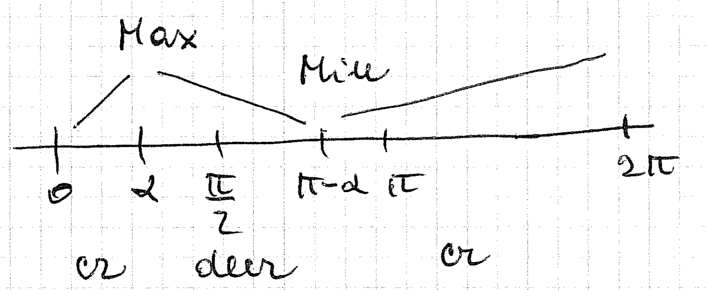


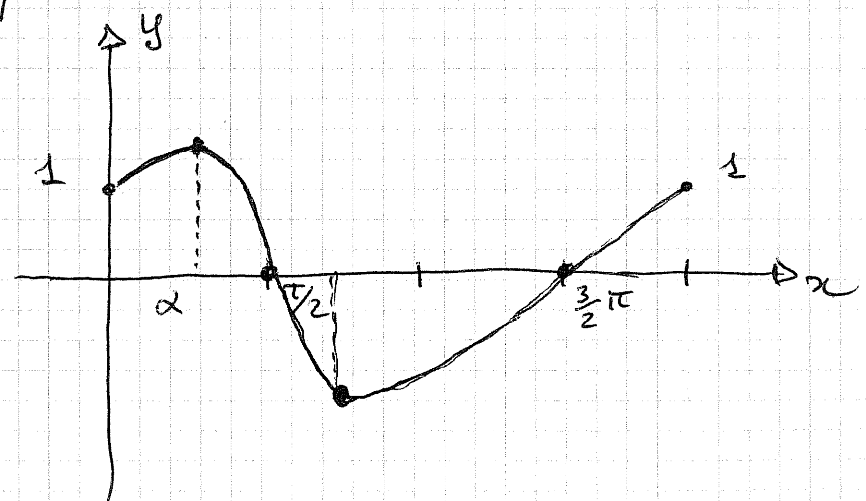
$y' = -\operatorname{sen} x \cdot e^{+\frac{1}{2}\operatorname{sen} x} + \cos x \cdot \frac{1}{2}\cos x \cdot e^{+\frac{1}{2}\operatorname{sen} x}$

$= -\frac{1}{2} e^{\frac{1}{2}\operatorname{sen} x} (\operatorname{sen}^2 x + 2\operatorname{sen} x - 1)$

$y' \geq 0$ $\operatorname{sen}^2 x + 2\operatorname{sen} x - 1 \leq 0$ $\cancel{-1-\sqrt{2} \leq} \operatorname{sen} x \leq \underbrace{\sqrt{2} - 1}_{\sim 0.4}$







$y = \sqrt{x^2-3x+2} - x$ (12) CE: $x^2-3x+2 \geq 0$ $\boxed{NOy''}$

$(-\infty, 1] [2, +\infty)$

$y(1) = -1$

$y(2) = -2$

$y(0) = \sqrt{2}$

$y = 0$ se $\sqrt{x^2-3x+2} = x$ $+3x = 2$ $x = 2/3$

$\lim_{x \to -\infty} y = +\infty$ $\sqrt{+\infty_2 - \infty_1 + 2} - (-\infty)$

$\infty_1 + \infty_1$

di 1° ordine
vediamo per As. OBL.

$\lim_{x \to -\infty} \frac{y}{x} = -2$ $\frac{\sqrt{x^2-3x+2} - x}{x}$ $\frac{|x|\sqrt{1-3/x+2/x^2}}{x} - 1$

(m) -1

$\lim_{x \to -\infty} (y+2x) = 3/2$ $\sqrt{x^2-3x+2} + x$

(q) $\underbrace{|x|}_{\text{ni } x(-\infty) \text{ è } -x} \sqrt{1+(2/x^2-3/x)} + x$

$y = -2x + \frac{3}{2}$

AS. OBL a $-\infty$

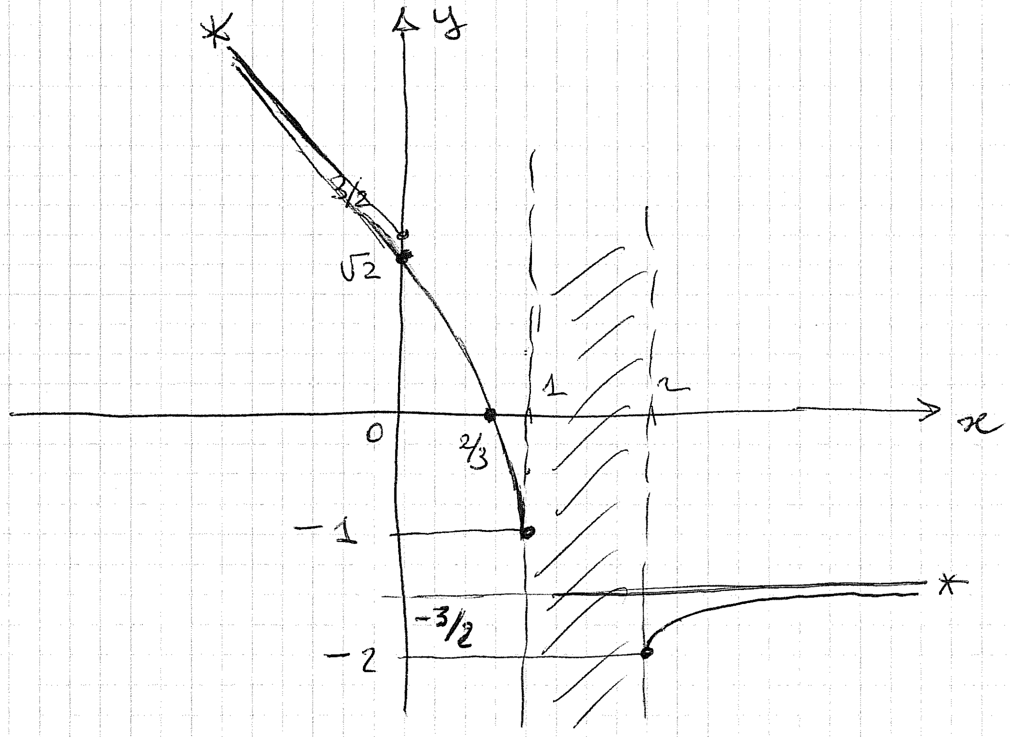
$$\underbrace{-x\Big[\underbrace{\left(1+\left(2/x^2-3/x\right)\right)^{1/2}-1}_{\sim (2/x^2-3/x)\cdot\frac{1}{2}}\Big]}_{-\left(\cancel{\frac{2}{x}}-3\right)\cdot\frac{1}{2} \quad \frac{3}{2}}$$

$\lim_{x \to +\infty} y = -\frac{3}{2}$ $+\infty - \infty$ $x\Big(\underbrace{\sqrt{1+2/x^2-3/x} - 1}_{\sim \frac{1}{2}\cdot\left(2/x^2 - 3/\cancel{x}\right)}\Big)$

$y = -3/2$

AS. ORIZZ.

$\frac{1}{x} - \frac{3}{2}$



$$y' = \frac{1}{2\sqrt{x^2-3x+2}} \cdot (2x-3) - 1 = \frac{2x-3 - 2\sqrt{x^2-3x+2}}{2\sqrt{x^2-3x+2}}$$

$y' \geq 0$ $\qquad 2x-3 \geq 2\sqrt{x^2-3x+2}$

se $x < 3/2$ ness soluz perché ($1°$ neg $\geq 2°$ pos)

$\Rightarrow$ $x < 3/2$ y decresce

se $x \geq \frac{3}{2}$ $\qquad 1°$ pos $\geq 2°$ pos ?

$$\cancel{4x^2} - \cancel{12x} + 9 \geq \cancel{4x^2} - \cancel{12x} + 8 \qquad \text{ok}$$

per $x \geq \frac{3}{2}$ y cresce

poiché la variazione decresce/cresce si ha ~~dopo~~ nelle due fasce distinte del CE non abbiamo massimi e/o minimi

osserviamo che

$$\left.\begin{array}{l} \lim\limits_{x \to 1^-} y' = -\infty \\ \lim\limits_{x \to 2^+} y' = +\infty \end{array}\right\} \text{ nei due punti di arresto la tangente è verticale}$$